

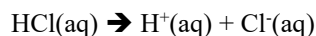
Intensive notes (Topic 4: Acids and bases)

Common acids and bases

Acid: A molecular compound which forms $\text{H}^+(\text{aq})$ as the only cation when dissolved in water

1. Hydrochloric acid (HCl)

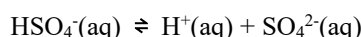
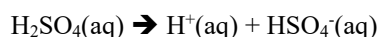
- A **strong monobasic** acid (There is 1 $\text{H}^+(\text{aq})$ can be produced by the complete ionization of one molecule of acid)



- It can be found in stomach juice. Concentrated HCl is volatile

2. Sulphuric acid (H_2SO_4)

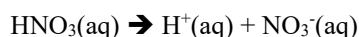
- A **strong dibasic** acid (There are 2 $\text{H}^+(\text{aq})$ can be produced by the complete ionization of one molecule of acid)



- Concentrated H_2SO_4 is hygroscopic, dehydrating and oxidizing

3. Nitric acid (HNO_3)

- A **strong monobasic** acid (There is 1 $\text{H}^+(\text{aq})$ can be produced by the complete ionization of one molecule of acid)



- Concentrated HNO_3 is volatile and oxidizing.
- Concentrated HNO_3 should be stored in brown bottle as HNO_3 can be decomposed by light. The acid often turns yellow due to the presence of NO_2 . The yellow color can be removed by bubbling air into the acid.

Chemical equation for the decomposition of HNO_3 : $4\text{HNO}_3 \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{NO}_2$

4. Ethanoic acid (CH_3COOH)

- A **weak monobasic** acid (There is 1 $\text{H}^+(\text{aq})$ can be produced by the complete ionization of one molecule of acid)



- It can be found in vinegar. Concentrated ethanoic acid is volatile

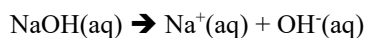
Common bases

Base: A compound which reacts with an acid forming salt and water only (Examples: Metal hydroxides, metal oxides, NH_3)

Alkalis: A base which is soluble in water (Examples: Hydroxides of Group I and some Group II metals, NH_3)

1. Sodium hydroxide (NaOH) and potassium hydroxide (KOH)

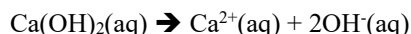
- Both are **strong** alkali



- Concentrated NaOH can be found in drain cleaner
- Concentrated NaOH and KOH are hygroscopic (absorb water) and absorb $\text{CO}_2(\text{g})$ in air readily

2. Calcium hydroxide ($\text{Ca}(\text{OH})_2$)

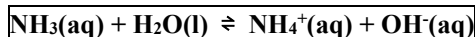
- A **strong** base, but only slightly soluble in water



- $\text{Ca}(\text{OH})_2(\text{aq})$ is also known as limewater. It can be used to neutralize acidic soils.

3. Aqueous ammonia (NH_3)

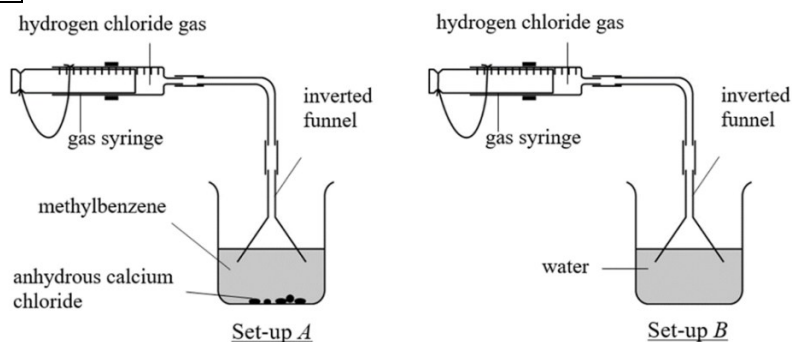
- A **weak** alkali



- NH_3 has a characteristic pungent smell. $\text{NH}_3(\text{aq})$ are volatile
- $\text{NH}_3(\text{g})$ can be tested by moist red litmus paper. It ionizes in water to give $\text{OH}^-(\text{aq})$ which turns red litmus blue.

Intensive notes (Topic 4: Acids and bases)

Water and acidic properties



Test	Set-up A (HCl dissolved in methylbenzene)	Set-up B (Hydrochloric acid)
Action on dry blue litmus paper	No observable change (H ⁺ ions are absent)	Changes from blue to red (H ⁺ ions are present)
Electrical conductivity	Does not conduct electricity (HCl remains as neutral molecules)	Conducts electricity (HCl ionizes in water to give H ⁺ and Cl ⁻)
Action on anhydrous Na ₂ CO ₃ (s)	No observable change (H ⁺ ions are absent)	Colorless gas bubbles given out (2H ⁺ (aq) + CO ₃ ²⁻ (aq) → CO ₂ (g) + H ₂ O(l))

- Use of inverted funnel can increase surface area for dissolving HCl(g), prevent sucking back of water.

Diluting concentrated acids / alkalis

- Safety precautions
 1. Wear safety goggles and protective gloves
 2. Dilution of volatile acids / alkalis (Conc. HCl, HNO₃, NH₃ etc.) should be performed in fume cupboard.
- Small amount of the concentrated acid / alkali should be added slowly to a large amount of water with stirring, but not vice versa. This is because large amount of heat will be produced during dilution. This can vaporize the water and cause splashing out of the acid /alkali

Basicity of common acids

Acid	Ionization in water	Basicity of acid
Hydrochloric acid Nitric acid Nitrous acid Ethanoic acid	HCl(aq) → H ⁺ (aq) + Cl ⁻ (aq) HNO ₃ (aq) → H ⁺ (aq) + NO ₃ ⁻ (aq) HNO ₂ (aq) ⇌ H ⁺ (aq) + NO ₂ ⁻ (aq) CH ₃ COOH(aq) ⇌ CH ₃ COO ⁻ (aq) + H ⁺ (aq)	1 (Monobasic)
Sulphuric acid Sulphurous acid Carbonic acid Oxalic acid	H ₂ SO ₄ (aq) → 2H ⁺ (aq) + SO ₄ ²⁻ (aq) H ₂ SO ₃ (aq) ⇌ 2H ⁺ (aq) + SO ₃ ²⁻ (aq) H ₂ CO ₃ (aq) ⇌ 2H ⁺ (aq) + CO ₃ ²⁻ (aq) H ₂ C ₂ O ₄ (aq) ⇌ 2H ⁺ (aq) + C ₂ O ₄ ²⁻ (aq)	2 (Dibasic)
Phosphoric acid Citric acid	H ₃ PO ₄ (aq) ⇌ 3H ⁺ (aq) + PO ₄ ³⁻ (aq) C ₆ H ₈ O ₇ (aq) ⇌ 3H ⁺ (aq) + C ₆ H ₅ O ₇ ³⁻ (aq)	3 (Tribasic)

Intensive notes (Topic 4: Acids and bases)

Reactions of acids and bases

	Acid	Metal	O ²⁻ / OH ⁻	CO ₃ ²⁻ / HCO ₃ ⁻	NH ₃	NH ₄ ⁺	CO ₂ / SO ₂	Products
1.	✓	✓						Salt + H ₂
2.	✓		✓					Salt + H ₂ O
3.	✓			✓				Salt + H ₂ O + CO ₂
4.	✓				✓			NH ₄ ⁺ salt
5.			✓			✓		Salt + H ₂ O + NH ₃
6.			✓				✓	CO ₃ ²⁻ / SO ₃ ²⁻ salt + H ₂ O

Reaction 1

Acid + Metal → Salt + Hydrogen

2HCl(aq) + Zn(s) → ZnCl ₂ (aq) + H ₂ (g)	Ionic equation: Zn(s) + 2H ⁺ (aq) → Zn ²⁺ (aq) + H ₂ (g)
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Reaction 2

Acid + Metal oxide → Salt + Water

Acid + Metal hydroxide → Salt + Water

2HCl(aq) + CuO(s) → CuCl ₂ (aq) + H ₂ O(l)	Ionic equation: 2H ⁺ (aq) + CuO(s) → Cu ²⁺ (aq) + H ₂ O(l)
3H ₂ SO ₄ (aq) + Al ₂ O ₃ (s) → Al ₂ (SO ₄) ₃ (aq) + 3H ₂ O(l)	Ionic equation: 6H ⁺ (aq) + Al ₂ O ₃ (s) → 2Al ³⁺ (aq) + 3H ₂ O(l)
2HNO ₃ (aq) + Ca(OH) ₂ (aq) → Ca(NO ₃) ₂ (aq) + 2H ₂ O(l)	Ionic equation: H ⁺ (aq) + OH ⁻ (aq) → H ₂ O(l)

Acid + Carbonate salt → Salt + Water + Carbon dioxide

Reaction 3

Acid + Hydrogencarbonate salt → Salt + Water + Carbon dioxide

Acid + Solid carbonate salt 2HCl(aq) + CaCO ₃ (s) → CaCl ₂ (aq) + H ₂ O(l) + CO ₂ (g)	Ionic equation: 2H ⁺ (aq) + CaCO ₃ (s) → Ca ²⁺ (aq) + H ₂ O(l) + CO ₂ (g)
Acid + Aqueous carbonate salt H ₂ SO ₄ (aq) + Na ₂ CO ₃ (aq) → Na ₂ SO ₄ (aq) + H ₂ O(l) + CO ₂ (g)	Ionic equation: 2H ⁺ (aq) + CO ₃ ²⁻ (aq) → H ₂ O(l) + CO ₂ (g)
Acid + Solid hydrogencarbonate salt HCl(aq) + NaHCO ₃ (s) → NaCl(aq) + H ₂ O(l) + CO ₂ (g)	Ionic equation: H ⁺ (aq) + NaHCO ₃ (s) → Na ⁺ (aq) + H ₂ O(l) + CO ₂ (g)
Acid + Aqueous hydrogencarbonate salt HNO ₃ (aq) + KHCO ₃ (aq) → KNO ₃ (aq) + H ₂ O(l) + CO ₂ (g)	Ionic equation: H ⁺ (aq) + HCO ₃ ⁻ (aq) → H ₂ O(l) + CO ₂ (g)

Reaction 4

Acid + Ammonia → Ammonium salt

2NH ₃ (aq) + H ₂ SO ₄ (aq) → (NH ₄) ₂ SO ₄ (aq)	Ionic equation: NH ₃ (aq) + H ⁺ (aq) → NH ₄ ⁺ (aq)
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Ammonium salt + Metal hydroxide → Salt + Ammonia + Water

Reaction 5

Ammonium salt + Metal oxide → Salt + Ammonia + Water

Solid ammonium salt + Base NH ₄ Cl(s) + NaOH(aq) → NaCl(aq) + NH ₃ (g) + H ₂ O(l)	Ionic equation: NH ₄ Cl(s) + OH ⁻ (aq) → Cl ⁻ (aq) + NH ₃ (g) + H ₂ O(l)
Aqueous ammonium salt + Base (NH ₄) ₂ SO ₄ (aq) + 2NaOH(aq) → Na ₂ SO ₄ (aq) + 2NH ₃ (g) + 2H ₂ O(l)	Ionic equation: NH ₄ ⁺ (aq) + OH ⁻ (aq) → NH ₃ (g) + H ₂ O(l)
2NH ₄ NO ₃ (s) + CaO(s) → Ca(NO ₃) ₂ (s) + 2NH ₃ (g) + 2H ₂ O(l)	Ionic equation: ---

Reaction 6

Carbon dioxide (Sulphur dioxide) + Metal hydroxide → Metal carbonate (Metal sulphite) + Water

2NaOH(aq) + CO ₂ (g) → Na ₂ CO ₃ (aq) + H ₂ O(l)	Ionic equation: 2OH ⁻ (aq) + CO ₂ (g) → CO ₃ ²⁻ (aq) + H ₂ O(l)
2KOH(aq) + SO ₂ (g) → K ₂ SO ₃ (aq) + H ₂ O(l)	Ionic equation: 2OH ⁻ (aq) + SO ₂ (g) → SO ₃ ²⁻ (aq) + H ₂ O(l)

Intensive notes (Topic 4: Acids and bases)

Precipitation (Salt + alkali)

Cation	Color	Addition of NaOH(aq) / KOH(aq)	Addition of NH ₃ (aq)
K ⁺	Colorless	No observable change	No observable change
Na ⁺	Colorless	No observable change	No observable change
Ca ²⁺	Colorless	White precipitate forms $\text{Ca}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Ca}(\text{OH})_2(\text{s})$	No observable change
Mg ²⁺	Colorless	White precipitate forms $\text{Mg}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$	White precipitate forms $\text{Mg}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$
Al ³⁺	Colorless	White precipitate forms and it dissolves in excess NaOH(aq) to give colorless solution $\text{Al}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Al}(\text{OH})_3(\text{s})$ $\text{Al}(\text{OH})_3(\text{s}) + \text{OH}^{-}(\text{aq}) \rightarrow \text{Al}(\text{OH})_4^{-}(\text{aq})$	White precipitate forms $\text{Al}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Al}(\text{OH})_3(\text{s})$
Zn ²⁺	Colorless	White precipitate forms and it dissolves in excess NaOH(aq) to give colorless solution $\text{Zn}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Zn}(\text{OH})_2(\text{s})$ $\text{Zn}(\text{OH})_2(\text{s}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Zn}(\text{OH})_4^{2-}(\text{aq})$	White precipitate forms and it dissolves in excess NH ₃ (aq) to give colorless solution $\text{Zn}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Zn}(\text{OH})_2(\text{s})$ $\text{Zn}(\text{OH})_2(\text{s}) + 4\text{NH}_3(\text{aq}) \rightarrow \text{Zn}(\text{NH}_3)_4^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq})$
Fe ²⁺	Green	Green precipitate forms $\text{Fe}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s})$	Green precipitate forms $\text{Fe}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s})$
Fe ³⁺	Yellow	Brown precipitate forms $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$	Brown precipitate forms $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$
Pb ²⁺	Colorless	White precipitate forms and it dissolves in excess NaOH(aq) to give colorless solution $\text{Pb}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Pb}(\text{OH})_2(\text{s})$ $\text{Pb}(\text{OH})_2(\text{s}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Pb}(\text{OH})_4^{2-}(\text{aq})$	White precipitate forms $\text{Pb}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Pb}(\text{OH})_2(\text{s})$
Cu ²⁺	Blue	Blue precipitate forms $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$	Blue precipitate forms and it dissolves in excess NH ₃ (aq) to give deep blue solution $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$ $\text{Cu}(\text{OH})_2(\text{s}) + 4\text{NH}_3(\text{aq}) \rightarrow \text{Cu}(\text{NH}_3)_4^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq})$
Ag ⁺	Colorless	Dark brown precipitate forms $2\text{Ag}^{+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Ag}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l})$	Dark brown precipitate forms and it dissolves in excess NH ₃ (aq) to give a colorless solution $2\text{Ag}^{+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Ag}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l})$ $\text{Ag}_2\text{O}(\text{s}) + 4\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Ag}(\text{NH}_3)_2^{+}(\text{aq}) + 2\text{OH}^{-}(\text{aq})$
NH ₄ ⁺	Colorless	Colorless pungent smell gas forms which can turn moist red litmus paper blue $\text{NH}_4^{+}(\text{aq}) + \text{OH}^{-}(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$	No observable change

Intensive notes (Topic 4: Acids and bases)

Precipitation (Salt + Salt / Acid)

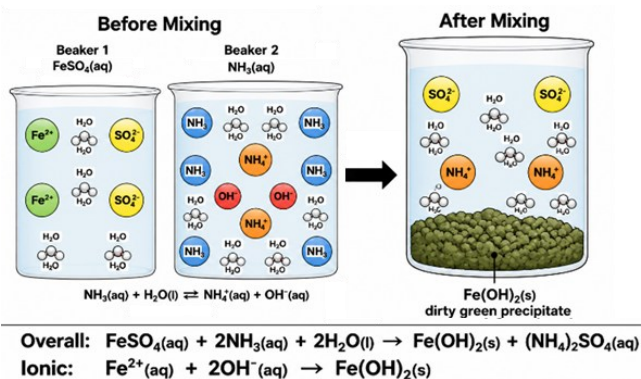
Compound of	Solubility in water
NH_4^+ , Na^+ , K^+ , NO_3^- , HCO_3^-	All are soluble
Cl^- , Br^- , I^-	All are soluble (Except AgCl , AgBr , AgI , PbCl_2 , PbBr_2 , PbI_2)
SO_4^{2-}	All are soluble (Except BaSO_4 , CaSO_4 , PbSO_4)
CO_3^{2-}	All are insoluble (Except Na_2CO_3 , K_2CO_3 , $(\text{NH}_4)_2\text{CO}_3$)
O^{2-} / OH^-	All are insoluble (Except Na_2O , K_2O , NaOH , KOH) CaO , MgO , $\text{Ca}(\text{OH})_2$ and $\text{Mg}(\text{OH})_2$ are <u>slightly soluble</u>

Consider the following examples:

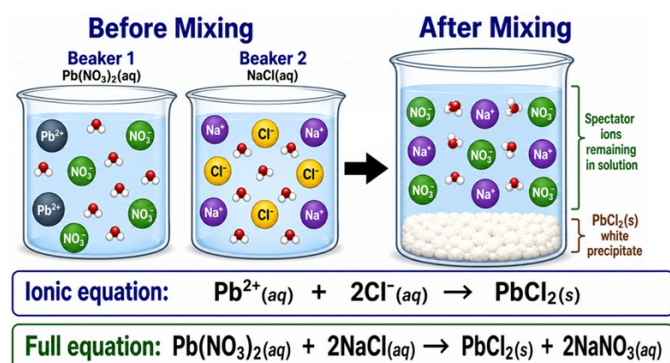
Solution A	Solution B	Chemical equation (A is added to B)
$\text{AgNO}_3(\text{aq})$	$\text{Cl}^-(\text{aq})$	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
$\text{Pb}(\text{NO}_3)_2(\text{aq})$	$\text{Cl}^-(\text{aq})$	$\text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{PbCl}_2(\text{s})$
$\text{Pb}(\text{NO}_3)_2(\text{aq})$	$\text{SO}_4^{2-}(\text{aq})$	$\text{Pb}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{PbSO}_4(\text{s})$
$\text{Ca}(\text{NO}_3)_2(\text{aq})$ / $\text{CaCl}_2(\text{aq})$ / $\text{Ca}(\text{OH})_2(\text{aq})$	$\text{SO}_4^{2-}(\text{aq})$	$\text{Ca}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{CaSO}_4(\text{s})$
$\text{Ba}(\text{NO}_3)_2(\text{aq})$ / $\text{BaCl}_2(\text{aq})$ / $\text{Ba}(\text{OH})_2(\text{aq})$	$\text{SO}_4^{2-}(\text{aq})$	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
$\text{Na}_2\text{CO}_3(\text{aq})$ / $\text{K}_2\text{CO}_3(\text{aq})$ / $(\text{NH}_4)_2\text{CO}_3(\text{aq})$	Aqueous salt [Except $\text{Na}^+(\text{aq})$, $\text{K}^+(\text{aq})$, $\text{NH}_4^+(\text{aq})$]	$2\text{M}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{M}_2\text{CO}_3(\text{s})$ $\text{M}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{MCO}_3(\text{s})$

Diagrams illustrating precipitation reaction:

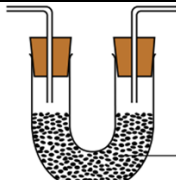
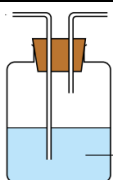
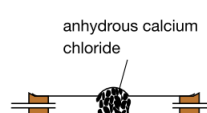
Example 1: $\text{FeSO}_4(\text{aq}) + \text{NH}_3(\text{aq})$



Example 2: $\text{Pb}(\text{NO}_3)_2(\text{aq}) + \text{NaCl}(\text{aq})$



Drying gas

		
Anhydrous CaO (Drying NH_3)	Concentrated H_2SO_4 (Drying acidic gases)	Anhydrous CaCl_2 (Drying acidic gases)

Intensive notes (Topic 4: Acids and bases)

Concentration calculation

$\text{Concentration} = \frac{\text{Amount of solute}}{\text{Volume of solution}}$	$\text{Molarity (mol dm}^{-3}\text{)} = \frac{\text{Mole (mol)}}{\text{Volume (dm}^3\text{)}}$
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Example 1 (Molarity calculation)

Solution X is prepared by dissolving 25.0 g of $\text{Fe}_2(\text{SO}_4)_3(\text{s})$ in 500.0 cm^3 water. (Formula mass of $\text{Fe}_2(\text{SO}_4)_3 = 399.9$)

- (a) Calculate the concentration of the solution, in g dm^{-3} .

Answer: Concentration (in g dm^{-3}) = $25 / 500 = 0.05 \text{ g dm}^{-3}$

- (b) Calculate the molarity of the solution.

Answer: Mole of $\text{Fe}_2(\text{SO}_4)_3 = 25 / 399.9 = 0.0625$

Molarity of $\text{Fe}_2(\text{SO}_4)_3 = 0.0625 / (500/1000) = 0.125 \text{ M}$

- (c) Calculate the concentration of $\text{Fe}^{3+}(\text{aq})$ in solution X.

Answer: Since 1 mole of $\text{Fe}_2(\text{SO}_4)_3$ can dissociate to give 2 moles of Fe^{3+}

Mole of $\text{Fe}^{3+}(\text{aq}) = 0.0625 \times 2 = 0.125$

Concentration of $\text{Fe}^{3+}(\text{aq}) = 0.125 / (500/1000) = 0.250 \text{ M}$

Example 2 (Dilution)

Solution X contains 100 cm^3 of 8 M $\text{HCl}(\text{aq})$

- (a) 400 cm^3 of distilled water is added to X. What is the concentration of HCl in the resulting mixture?

Answer: $(M_1)(V_1) = (M_2)(V_2)$

$(8)(100) = (M_2)(100 + 400)$

$M_2 = 1.6 \text{ M}$

- (b) What volume of water is required to dilute 100 cm^3 of 8 M HCl to a concentration of 2 M?

Answer: $(M_1)(V_1) = (M_2)(V_2)$

$(8)(100) = (2)(100 + v)$

$v = 300 \text{ cm}^3$

Example 3 (Mixing two solutions with different concentrations of solute)

Solution X contains 200.0 cm^3 of 1.0 M $\text{Na}_2\text{SO}_4(\text{aq})$.

- (a) 300.0 cm^3 of 1.5 M $\text{NaCl}(\text{aq})$ is added to X. Calculate the concentration of $\text{Na}^+(\text{aq})$ in the resulting mixture.

Answer: $(M_1)(V_1) + (M_2)(V_2) = (M_3)(V_3)$

$(1.0 \times 2)(200/1000) + (1.5)(300/1000) = (M_3)(200/1000 + 300/1000)$

$M_3 = 1.7 \text{ M}$

*Since 1 mol N

- (b) What volume of 0.1 M $\text{Na}_3\text{PO}_4(\text{aq})$ is required to prepare a mixture containing 0.64 M $\text{Na}^+(\text{aq})$?

Answer: $(M_1)(V_1) + (M_2)(V_2) = (M_3)(V_3)$

$(200/1000)(1.0 \times 2) + (0.1 \times 3)(v) = (0.64)(200/1000 + v)$

$v = 0.8 \text{ dm}^3$

Intensive notes (Topic 4: Acids and bases)

Example 4 (Expressing concentration in different units)

- (a) The molarity of a sample of $\text{CaCl}_2(\text{aq})$ is 0.5 M. Express the concentration of CaCl_2 in g cm^{-3} .

(Formula mass of $\text{CaCl}_2 = 111.1$)

Answer:

$$\frac{\text{mol}}{\text{dm}^3} \xrightarrow[\times 1000]{\times \text{Molar mass}} \frac{\text{g}}{\text{cm}^3}$$

$$\text{Concentration in } \text{g cm}^{-3} = (0.5 \text{ mol} \times 111.1 \text{ g mol}^{-1}) / (1 \text{ dm}^3 \times 1000 \text{ cm}^3 \text{ dm}^{-3}) = 0.05555 \text{ g cm}^{-3}$$

- (b) The concentration of $\text{NaCl}(\text{aq})$ is 0.242 g cm^{-3} . Calculate the molarity of NaCl of this solution.

(Formula mass of $\text{NaCl} = 58.5$)

Answer:

$$\frac{\text{g}}{\text{cm}^3} \xrightarrow[\div 1000]{\div \text{Molar mass}} \frac{\text{mol}}{\text{dm}^3}$$

$$\text{Molarity (must be in } \text{mol dm}^{-3}) = (0.242 \text{ g} \div 58.5 \text{ g mol}^{-1}) / (1 \text{ dm}^3 \div 1000 \text{ cm}^3 \text{ dm}^{-3}) = 4.14 \text{ M}$$

Example 5 (Calculate the molarity of solution from density and percentage by mass of solute)

A sample of HCl has a density of 1.18 g cm^{-3} and the percentage by mass of HCl in this sample is 36%. Calculate the concentration of HCl in mol dm^{-3} . (Molecular mass of $\text{HCl} = 36.5$)

Answer: Take 1000 cm^3 for calculation Mass of HCl in 1000 cm^3 sample $= 1000 \times 1.18 \times 36\% = 424.8 \text{ g}$ Mole of HCl in 1000 cm^3 sample $= 424.8 / 36.5 = 11.6$ Concentration of $\text{HCl} = 11.6 / (1000/1000) = 11.6 \text{ M}$	In $v \text{ cm}^3$ solution: $\text{Molarity} = \frac{\frac{(\text{volume in } \text{cm}^3)(\text{density in } \text{g cm}^{-3})(\%) }{(\text{Molar mass})}}{\frac{(\text{volume in } \text{cm}^3)}{1000}}$
--	--

Color of indicators

1. Methyl orange

Color	Red	Orange	Yellow
pH range	< 3.1	3.1 – 4.4	> 4.4

2. Phenolphthalein

Color	Colorless	Very pale pink	Pink
pH range	< 8.3	8.3 – 10	> 10

3. Litmus

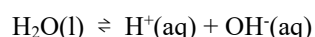
Color	Red	Purple	Blue
pH range	< 5	5 – 8	> 8

Note: The above indicators do not show the pH value of the solution. To measure the pH of the solution, we can use universal indicator or pH meter (more accurate)

Acidity and alkalinity of an aqueous solution

Acidic solution	Neutral solution	Alkaline solution
$[\text{H}^+(\text{aq})] > [\text{OH}^-(\text{aq})]$	$[\text{H}^+(\text{aq})] = [\text{OH}^-(\text{aq})]$	$[\text{H}^+(\text{aq})] < [\text{OH}^-(\text{aq})]$

Note: $[\text{H}^+(\text{aq})]$ and $[\text{OH}^-(\text{aq})]$ are always present in aqueous solution. That is, there are still $\text{H}^+(\text{aq})$ ions in a highly alkaline solution and there are still $\text{OH}^-(\text{aq})$ ions in a highly acidic solution.



Intensive notes (Topic 4: Acids and bases)

pH calculation

When calculating pH from given $[H^+(aq)]$:	When calculating $[H^+(aq)]$ from given pH:
$pH = -\log [H^+(aq)]$	$[H^+(aq)] = 10^{-pH}$

Example 1

- (a) Calculate the pH of 0.1 M HCl(aq)

Answer: $pH = -\log[H^+(aq)] = -\log(0.1) = 1$

- (b) Calculate the pH of 0.1 M $H_2SO_4(aq)$

Answer: $pH = -\log[H^+(aq)] = -\log(0.1 \times 2) = 0.699$

Example 2

- (a) The pH of a sample of HCl(aq) is 1.5. Calculate the concentration of HCl(aq)

Answer: $[H^+(aq)] = 10^{-pH} = 10^{-1.5} = 0.0316 \text{ M}$

Since $[H^+(aq)] : [HCl(aq)] = 1 : 1$, $[HCl(aq)] = 0.0316 \text{ M}$

- (b) The pH of a sample of $H_2SO_4(aq)$ is 1.5. Calculate the concentration of $H_2SO_4(aq)$

Answer: $[H^+(aq)] = 10^{-pH} = 10^{-1.5} = 0.0316 \text{ M}$

Since $[H^+(aq)] : [H_2SO_4(aq)] = 2 : 1$, $[H_2SO_4(aq)] = 0.0316 \times 1/2 = 0.0158 \text{ M}$

Example 3

It is given that at 25°C, $[H^+(aq)][OH^-(aq)] = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Calculate the pH of 0.01 M NaOH(aq)

Answer: $[H^+(aq)](0.01) = 1.0 \times 10^{-14}$

$[H^+(aq)] = 1.0 \times 10^{-12}$

$pH = -\log[H^+(aq)] = -\log(1.0 \times 10^{-12}) = 12$

- (b) Calculate the pH of 0.01 M $Ca(OH)_2(aq)$

Answer: $[H^+(aq)](0.01 \times 2) = 1.0 \times 10^{-14}$

$[H^+(aq)] = 5.0 \times 10^{-13}$

$pH = -\log[H^+(aq)] = -\log(5.0 \times 10^{-13}) = 12.3$

Example 4

It is given that at 25°C, $[H^+(aq)][OH^-(aq)] = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) The pH of a sample of KOH(aq) is 12.2. Calculate the concentration of KOH(aq).

Answer: $[H^+(aq)] = 10^{-pH} = 10^{-12.2} = 6.31 \times 10^{-13} \text{ M}$

$(6.31 \times 10^{-13})[OH^-(aq)] = 1.0 \times 10^{-14}$

$[OH^-(aq)] = 0.0158 \text{ M}$

Since $[OH^-(aq)] : [KOH(aq)] = 1 : 1$, $[KOH(aq)] = 0.0158 \text{ M}$

- (b) The pH of a sample of $Ca(OH)_2$ is 11.8. Calculate the concentration of $Ca(OH)_2(aq)$

Answer: $[H^+(aq)] = 10^{-pH} = 10^{-11.8} = 1.58 \times 10^{-12} \text{ M}$

$(1.58 \times 10^{-12})[OH^-(aq)] = 1.0 \times 10^{-14}$

$[OH^-(aq)] = 6.31 \times 10^{-3} \text{ M}$

Intensive notes (Topic 4: Acids and bases)

Since $[\text{OH}^-(\text{aq})] : [\text{Ca}(\text{OH})_2(\text{aq})] = 2 : 1$, $[\text{Ca}(\text{OH})_2(\text{aq})] = 6.31 \times 10^{-13} \times 1/2 = 3.15 \times 10^{-3} \text{ M}$

Strength of acid and alkali

Strong acid: An acid which ionizes completely in water	Strong alkali: An alkali which dissociates completely in water
HCl, HNO ₃ , H ₂ SO ₄	*NaOH, *KOH, *Ca(OH) ₂
Weak acid: An acid which ionizes slightly in water	Weak alkali: An alkali which ionizes slightly in water
#CH ₃ COOH, #H ₂ C ₂ O ₄ , H ₂ CO ₃ , H ₂ SO ₃ , H ₃ PO ₄	NH ₃

*Ions are present in NaOH(s), KOH(s) and Ca(OH)₂(s) already. They dissociate (not ionize) in water.

#Most organic acids are weak acids.

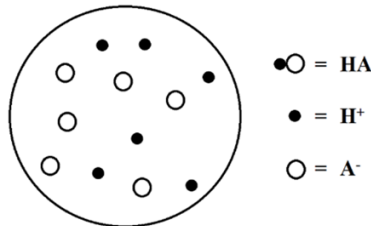
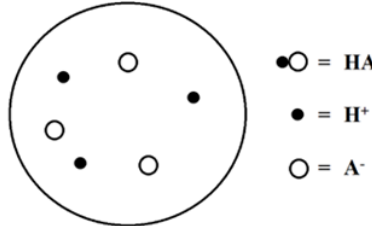
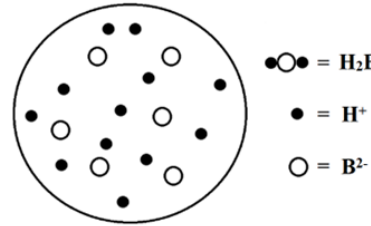
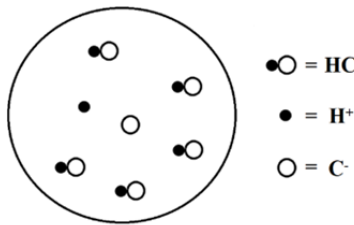
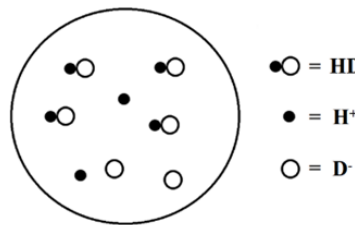
Ionization of strong acid HCl: $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$

Ionization of weak acid CH₃COOH: $\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$

Ionization of weak alkali NH₃: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$

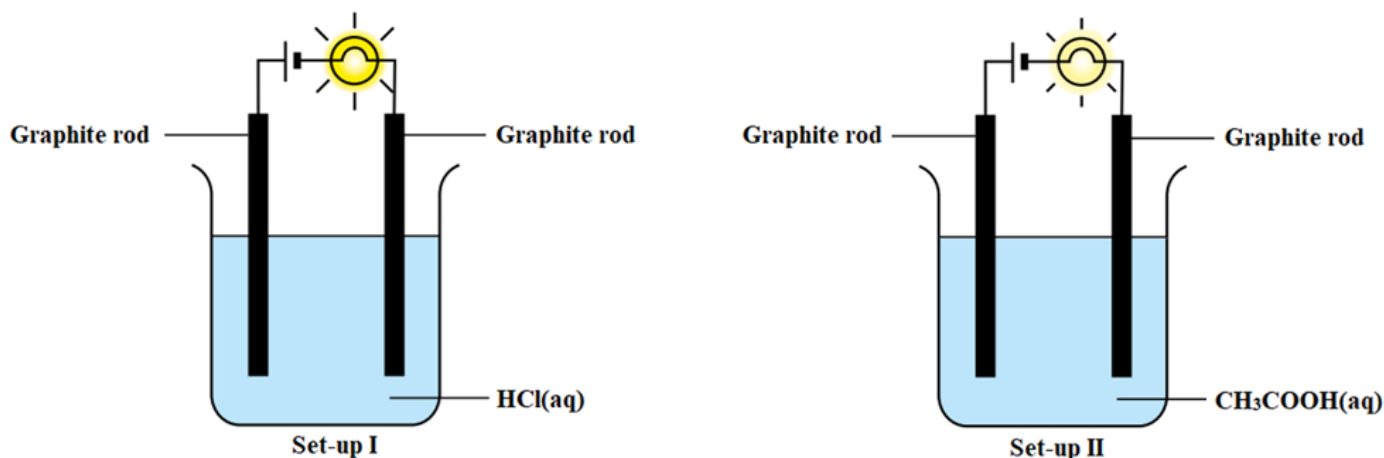
	100.0 cm ³ of 0.1 M HCl(aq)	100.0 cm ³ of 0.1 M CH ₃ COOH(aq)
Conductivity	Higher (as it contains higher concentration of mobile ions)	Lower (as it contains lower concentration of mobile ions)
pH	= 1.0 (As $[\text{H}^+(\text{aq})] = [\text{HCl}(\text{aq})]$)	> 1.0 (As $[\text{H}^+(\text{aq})] < [\text{CH}_3\text{COOH}(\text{aq})]$)
Mole of NaOH(aq) required for complete neutralization	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ 0.01 mol of NaOH required (As it contains 0.01 mole HCl)	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$ 0.01 mol of NaOH required (As it contains 0.01 mole CH ₃ COOH)
Temperature rise when 100.0 cm³ of 0.1 M NaOH(aq) is added	Higher	Lower (heat is absorbed for ionization of weak acid)

Summary on strength, concentration and basicity

 ●○ = HA ● = H⁺ ○ = A⁻	 ●○ = HA ● = H⁺ ○ = A⁻	 ●●○ = H₂B ● = H⁺ ○ = B²⁻
1.0 M HA(aq) (A strong monobasic acid)	0.5 M HA(aq) (A strong monobasic acid)	1.0 M H ₂ B(aq) (A strong dibasic acid)
 ●○ = HC ● = H⁺ ○ = C⁻	 ●○ = HD ● = H⁺ ○ = D⁻	
1.0 M HC(aq) (A weak monobasic acid)	1.0 M HD(aq) (A weak monobasic acid but stronger than HC)	

Experiment comparing strengths of acids and alkalis

Experiment 1



Observation: Set-up I has a higher electrical conductivity (and the light bulb glows brighter)

Explanation: HCl is a stronger acid than CH₃COOH(aq). HCl ionizes completely in water while CH₃COOH(aq) ionizes only slightly in water. Solution in set-up I has a higher concentration of mobile ions.

Fair comparison: Both HCl(aq) and CH₃COOH(aq) should have same volume, concentration and temperature.

Experiment 2

- (a) HA is a monobasic acid. At room conditions, the pH of a 0.05 M of HA(aq) is 1.30. Based on this information and appropriate calculation, comment whether HA is completely ionized in water.

Answer: Assume HA is a strong acid. $\text{HA} \rightarrow \text{H}^+(\text{aq}) + \text{A}^-(\text{aq})$

$$[\text{HA}(\text{aq})] : [\text{H}^+(\text{aq})] = 1 : 1$$

$$[\text{H}^+(\text{aq})] = 0.05 \text{ M}$$

$$\text{pH} = -\log[\text{H}^+(\text{aq})] = -\log(0.05) = 1.3$$

As the pH of the solution agrees with the theoretical value, it is a strong acid.

- (b) H₂B is a dibasic acid. At room conditions, the pH of a 0.005 M of H₂B(aq) is 4.1. Based on this information and appropriate calculation, comment whether H₂B is completely ionized in water.

Answer: Assume H₂B is a strong acid. $\text{H}_2\text{B} \rightarrow 2\text{H}^+(\text{aq}) + \text{B}^{2-}(\text{aq})$

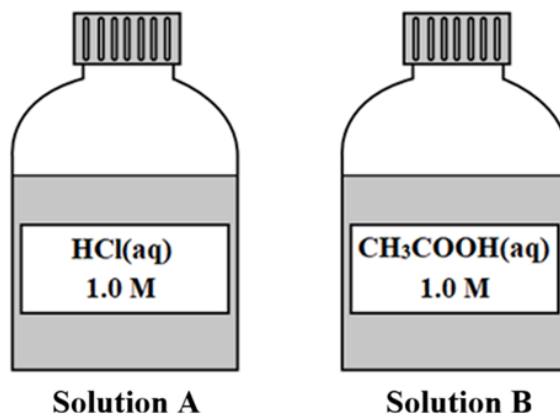
$$[\text{H}_2\text{B}(\text{aq})] : [\text{H}^+(\text{aq})] = 1 : 2$$

$$[\text{H}^+(\text{aq})] = 0.005 \times 2 = 0.01 \text{ M}$$

$$\text{pH} = -\log[\text{H}^+(\text{aq})] = -\log(0.01) = 2 < 4.1$$

As the pH of the solution is higher than the theoretical value, it is a weak acid.

Experiment 3

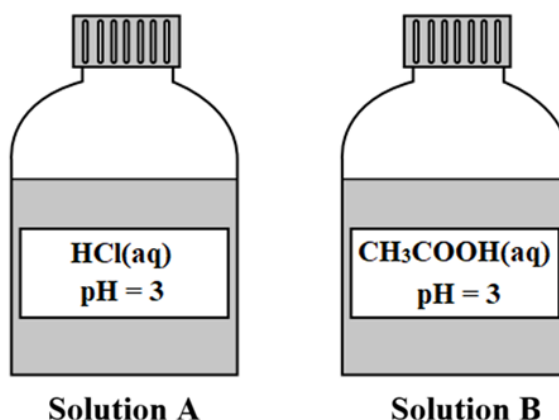


Deduce whether solution A or solution B has a lower pH.

Answer: Solution A has a lower pH. HCl(aq) is a strong acid which completely ionizes in water to give $\text{H}^+(\text{aq})$. $\text{CH}_3\text{COOH(aq)}$ is a weak acid which slightly ionizes in water to give $\text{H}^+(\text{aq})$. Since solution A has a higher $[\text{H}^+(\text{aq})]$, it should have a lower pH.

Experiment 4

Consider the following two acids



(a) Deduce whether HCl(aq) or $\text{CH}_3\text{COOH(aq)}$ has a higher molarity.

Answer: For HCl(aq) , $[\text{H}^+(\text{aq})] = [\text{Cl}^-(\text{aq})] = 0.001 \text{ M}$. As HCl ionizes completely in water, $[\text{HCl(aq)}]$ should be 0.001 M . For $\text{CH}_3\text{COOH(aq)}$, $[\text{H}^+(\text{aq})] = [\text{CH}_3\text{COO}^-(\text{aq})] = 0.001 \text{ M}$. As CH_3COOH ionizes slightly in water, $[\text{CH}_3\text{COOH(aq)}]$ must be higher than $[\text{H}^+(\text{aq})]$ (0.001 M). Therefore CH_3COOH should have a higher molarity.

(b) Solutions A and B are diluted by 10 times. State and explain the change in pH of the two solutions.

Answer: When solution A (HCl , a strong acid) is diluted by 10 times, $[\text{H}^+(\text{aq})]$ decreases by 10 times and the pH of the resultant solution increases by 1 (i.e. 4).

When solution C (CH_3COOH , a weak acid) is diluted by 10 times, besides the dilution of the $\text{H}^+(\text{aq})$ originally present, some of the unionized ethanoic acid will ionize to give additional $\text{H}^+(\text{aq})$.

Hence, $[\text{H}^+(\text{aq})]$ in the resultant solution will be greater than 10^{-4} M and has a pH smaller than 4.0.

Intensive notes (Topic 4: Acids and bases)

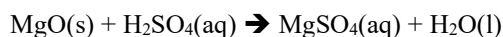
Soluble salt preparation

Excess metal oxide (or metal hydroxide / metal carbonate / metal) + acid $\rightarrow$ Salt + Water

Excess metal A + Metal B salt $\rightarrow$ Metal B + Metal A Salt

Example: Preparation of magnesium sulphate crystals

Step 1*. Add excess magnesium oxide to dilute sulphuric acid, then filter the mixture to obtain magnesium sulphate solution



Step 2^ Heat the magnesium sulphate solution until it becomes saturated

Step 3# Cool the solution slowly to obtain the crystals. Filter to obtain the crystals.

Step 4@ Wash the crystals with small amount of cold distilled water

Step 5% Dry the crystals with filter paper

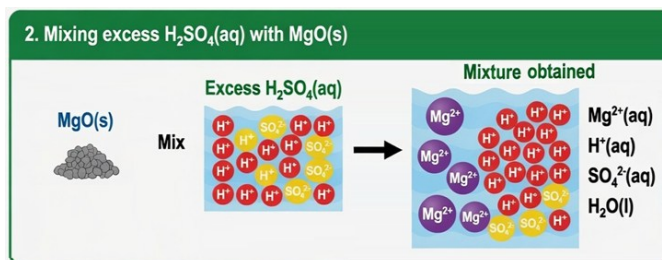
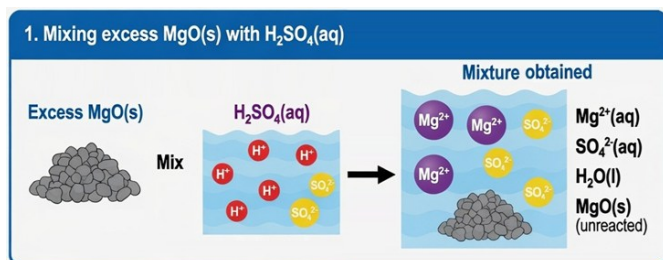
* Excess MgO can ensure that all the sulphuric acid has reacted and the product is not contaminated by the acid.

^ To test whether the solution is saturated, we can dip a glass rod into the solution and take it out. If crystals are formed on it upon cooling, the solution is saturated

When the temperature of the solution drops, the solubility will decrease. Water cannot hold all the dissolved solute and some of the salt crystals will be formed

@ Washing the crystals can remove the water-soluble impurities. Using a small amount can reduce loss of the salt.

% It is not suitable to dry the crystals by heating. Otherwise water of crystallization may be removed / Salt may be decomposed upon heating.



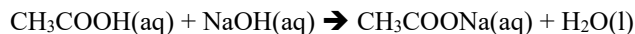
Group I / ammonium salt preparation

Acid + Alkali $\rightarrow$ Salt + Water

Acid + Metal carbonate $\rightarrow$ Salt + Water + Carbon dioxide

Example: Preparation of sodium ethanoate crystals

Step 1 Place a fixed volume of sodium hydroxide solution in a conical flask. Add a few drops of phenolphthalein. Run the ethanoic acid from the burette into the conical flask until the solution in the flask turns from pink to colorless



Step 2 Repeat step 1 with exactly the same volumes of acid and alkali used. However DO NOT add phenolphthalein this time

Step 3. Heat the sodium ethanoate solution until it becomes saturated

Step 4. Cool the solution slowly to obtain the crystals. Filter to obtain the crystals.

Step 5. Wash the crystals with small amount of cold distilled water

Step 6. Dry the crystals with filter paper

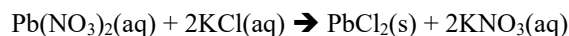
Intensive notes (Topic 4: Acids and bases)

Insoluble salt preparation

[Cation] nitrate solution + Potassium / sodium [anion] solution → Insoluble salt + Potassium / sodium nitrate solution

Example: Preparation of lead(II) chloride

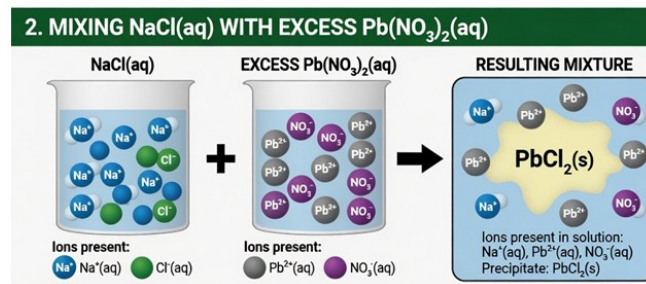
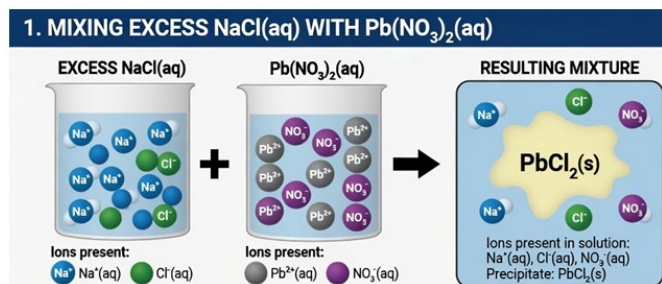
Step 1. Mix lead(II) nitrate solution and potassium chloride solution



Step 2. Filter to obtain the precipitate (as residue)

Step 3. Wash the precipitate with large amount of distilled water.

Step 4. Dry the precipitate with filter paper



Applications of neutralization

1. Adjusting soil pH

Acidic soil can be neutralized by slaked lime (Ca(OH)₂) or quicklime (CaO)

Alkaline soil can be neutralized by ammonium sulphate (which is also a fertilizer)

2. Neutralizing excess acid in stomach (antacids)

Mg(OH)₂ and Al(OH)₃ are weak bases in antacids. (Strong base such as NaOH are not used as it is too corrosive)

CaCO₃ and NaHCO₃ can also be used. However, CO₂(g) produced can lead to uncomfortable feeling in stomach

3. Treating industrial waste

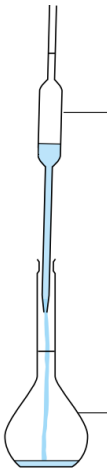
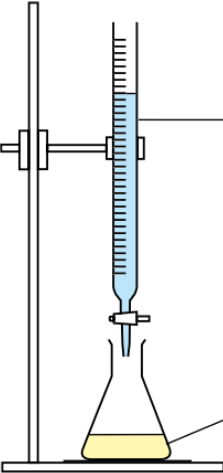
Acidic liquid waste should be treated with slaked lime (Ca(OH)₂) or sodium carbonate

Alkaline liquid waste should be treated with dilute sulphuric acid

Note: Neutralization is NOT an application in acid / alkali burns. This is because neutralization is an exothermic reaction (a process which releases heat). In case of acid / alkali burns, put the affected area under cool running water.

Intensive notes (Topic 4: Acids and bases)

Titration

Dilution set-up		Titration set-up	
	Pipette - Rinse with distilled water and then the solution it is going to deliver - It is used to deliver a specific volume of a solution accurately.		*Burette - Rinse with distilled water and then the solution it is going to deliver - It is used to deliver variable volumes (ranging from 0.05 to 50.00 cm³) of a solution accurately
	Volumetric flask - Rinse with distilled water - It is used to make up a solution to a specific volume accurately.		Conical flask - Rinse with distilled water - It is used to hold the solution to be titrated.

* While preparing a burette filled with solution, make sure that:

1. The space between the tap and the tip is filled with the solution
2. No air bubbles inside the burette
3. The meniscus is not above the zero mark
4. The burette is in a vertical position

Standard solution preparation

Standard solution is a solution with accurately known concentration

Primary standard is a substance which can be used to prepare a standard solution directly. It can be weighed out accurately and used to prepare a solution of accurate concentration.

Standard solution can be used to determine the concentration / molar mass / percentage by mass of another reagent etc.

Properties of primary standard:

- Readily available in high purity
- Chemically stable
- Highly soluble in water
- Does not absorb water vapor
- Does not lose water of crystallization when hydrated
- Non-volatile and non-toxic

Example 1: Preparation of 250.0 cm³ standard 1.0 M Na₂CO₃(aq)

Mass of Na₂CO₃(s) needed = (1.0)(250/1000)(106) = 26.5 g

- Step 1 Dissolve 26.5 g of Na₂CO₃(s) in a sufficient amount of distilled water in a beaker
- Step 2 Transfer the solution into a 250.0 cm³ volumetric flask
- Step 3 Rinse the beaker with distilled water several times and transfer all the washings into the volumetric flask
- Step 4 Make up to the graduation mark with distilled water

Example 2: Preparation of standard 100.0 cm³ standard 0.3 M NaOH(aq) from standard 1.2 M NaOH(aq)

Volume of 1.2 M NaOH(aq) needed = (0.3 / 1.2)(100) = 25 cm³

- Step 1 Use a 25.0 cm³ pipette to transfer 25.0 cm³ of the 1.2 M NaOH(aq) to the 100.0 cm³ volumetric flask.
- Step 2 Make up to the graduation mark with distilled water

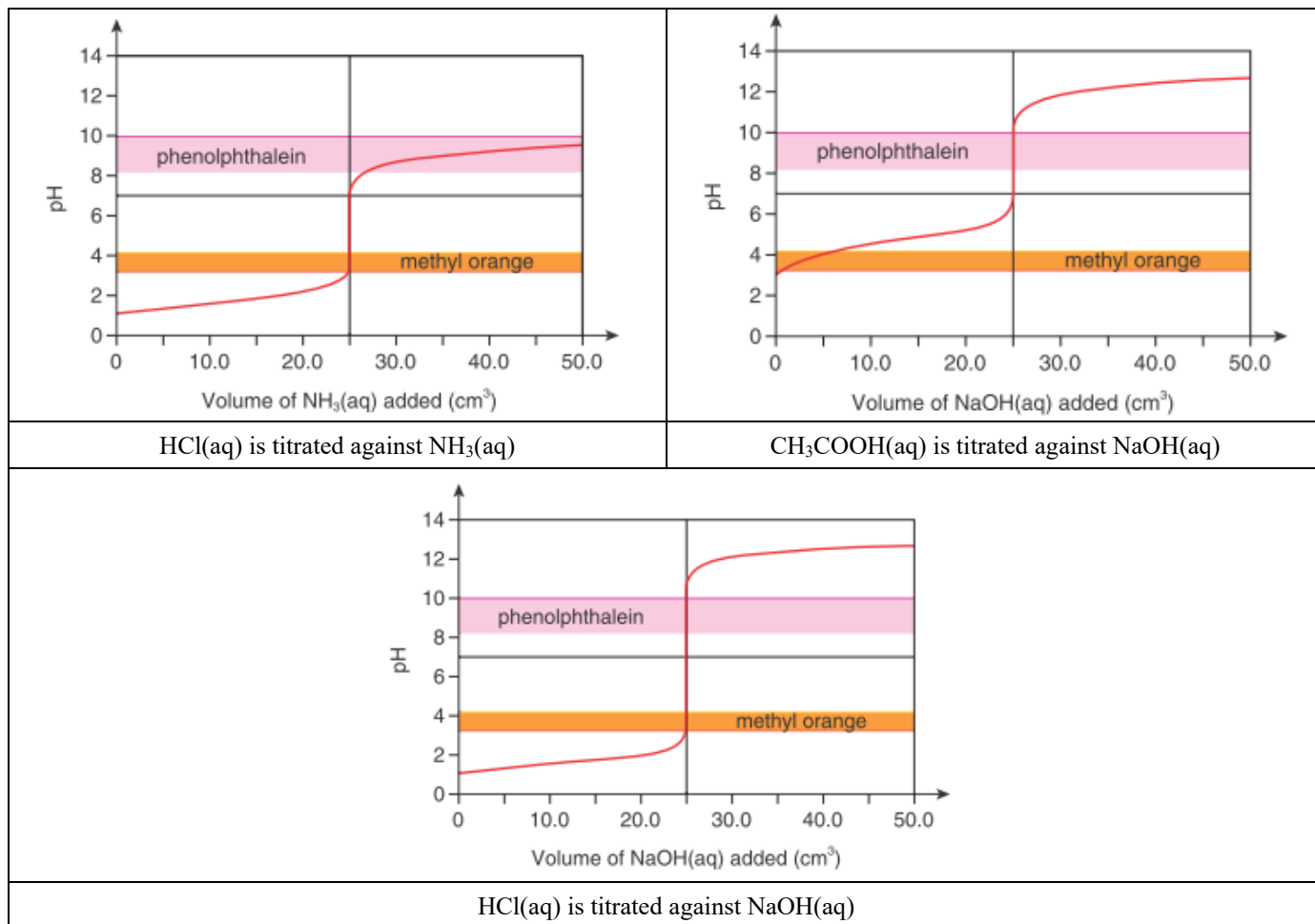
Intensive notes (Topic 4: Acids and bases)

pH curve and indicator

Equivalence point is the point at which an acid and an alkali have just completely reacted with each other.

End point is the point at which the indicator changes colour sharply.

The following titration curve



Choice of indicator

Titration between:	Strong acid	Weak acid
Strong base	Phenolphthalein and methyl orange	Phenolphthalein
Weak base	Methyl orange	No suitable indicator

*Methyl orange changes its color at around pH 3-5. Phenolphthalein changes its color at around pH 8-10.

Color change at end point

	Acidic solution in conical flask	Basic solution in conical flask (including CO ₃ ²⁻ (aq) and HCO ₃ ⁻ (aq) salt)
Methyl Orange	From red to orange	From yellow to orange
Phenolphthalein	From colorless to pale pink	From pink to very pale pink

*Litmus solution is not used as indicator in titration as its colour change at end point is not sharp

Intensive notes (Topic 4: Acids and bases)

Titration calculations

$\text{Concentration} = \frac{\text{Mole}}{\text{Volume}}$	$\text{Basicity} = \frac{\text{Mole of alkali}}{\text{Mole of acid}}$
$\text{Molar mass} = \frac{\text{Mass}}{\text{Mole}}$	$\text{Percentage by mass} = \frac{(\text{Mole})(\text{Molar mass})}{\text{Total mass of sample}} \times 100\%$

Example 1 (Calculate the concentration of solution)

25.0 cm³ of NH₃(aq) sample is diluted to 100.0 cm³ with distilled water. 25.00 cm³ of the diluted solution is titrated against 0.012 M H₂SO₄(aq). The table below shows the results of the titrations:

	Trial	1	2	3	4
Final burette reading / cm³	15.00	12.80	13.90	14.20	12.40
Initial burette reading / cm³	2.00	1.50	1.00	3.00	1.00

- (a) Calculate the reasonable average of the 0.012 M H₂SO₄(aq) used

Answer: $[(12.80 - 1.50) + (14.20 - 3.00) + (12.40 - 1.00)] / 3 = 11.30 \text{ cm}^3$ (2 decimal places + unit!!!)
(Trial 2 is not reasonable, should not be included)

- (b) Suggest a suitable indicator and state the color change at end point

Answer: Methyl orange. (Strong acid vs weak base, only methyl orange can be used)
From yellow to orange (11.30 cm³ H₂SO₄ → burette contains H₂SO₄ → conical flask contains NH₃)

- (c) Calculate the concentration of NH₃ sample

Answer: $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$
 $n_{\text{H}_2\text{SO}_4} = (11.30/1000) \times 0.012 = 1.356 \times 10^{-4}$
 $n_{\text{NH}_3} \text{ in } 25.00 \text{ cm}^3 \text{ diluted solution} = 1.356 \times 10^{-4} \times 2 = 2.712 \times 10^{-4}$
 $n_{\text{NH}_3} \text{ in } 100.0 \text{ cm}^3 \text{ diluted solution} = 2.712 \times 10^{-4} \times 100/25 = 1.0848 \times 10^{-3} = n_{\text{NH}_3} \text{ in } 25.0 \text{ cm}^3 \text{ sample}$
 Concentration of NH₃ = $1.0848 \times 10^{-3} / (25/1000) = 0.0433 \text{ M}$

- (d) Why was the titration repeated several times?

Answer: To reduce errors of the titration results.

Example 2 (Calculate the concentration of solution)

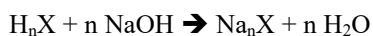
To find out the concentration of HCl, a laboratory technician first drew from the bottle a sample of 10.00 cm³ of the HCl and diluted it to 100.0 cm³ in a volumetric flask. The diluted acid sample was then used to titrate a standard Na₂CO₃(aq) placed in a conical flask. 10.00 cm³ of 1.06 mol dm⁻³ Na₂CO₃(aq) required 20.30 cm³ of the diluted acid sample to reach the end point. Using the titration result, calculate the concentration, in mol dm⁻³, of the HCl sample.

Answer: $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$
 $n_{\text{Na}_2\text{CO}_3} = (10/1000) \times 1.06 = 0.0106$
 $n_{\text{HCl}} \text{ in } 20.30 \text{ cm}^3 \text{ diluted sample} = 0.0106 \times 2 = 0.0212$
 $n_{\text{HCl}} \text{ in } 100.0 \text{ cm}^3 \text{ diluted sample} = 0.0212 \times 100/20.3 = 0.1044 = n_{\text{HCl}} \text{ in } 10.00 \text{ cm}^3 \text{ sample}$
 Concentration of HCl = $0.1044 / (10/1000) = 10.44 \text{ M}$

Example 3 (Calculate the basicity of acid)

24.62 cm³ of 0.207 M NaOH(aq) can completely neutralize 25.00 cm³ of 0.102 M tartronic acid. Calculate the basicity of tartronic acid.

Answer: Let H_nX be the formula of tartronic acid where n is the basicity of acid.



Mole of alkali : Mole of acid = n : 1

$$[(24.62/1000) \times 0.207] : [(25/1000) \times 0.102] = n : 1$$

$$n = 2 \quad (\text{Tartronic acid is a dibasic acid})$$

Example 4 (Calculate the basicity of acid)

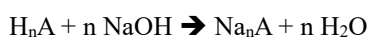
To determine the basicity of an unknown acid X (formula mass = 90.0), 1.8 g of acid X is dissolved in distilled water and made up to a 250.0 cm³ solution. 25.0 cm³ portions of the solution are then titrated against 0.22 M NaOH(aq). The mean titre required to reach the end point is 18.20 cm³. Calculate the basicity of acid X.

Answer: Let H_nA be the formula of acid X where n is the basicity of acid.

$$^n\text{NaOH used} = 0.22 \times 18.2/1000 = 0.004$$

$$\text{Total } ^n\text{X used} = 1.8/90 = 0.02$$

$$^n\text{X used in titration} = 0.02 \times 25/250 = 0.002$$



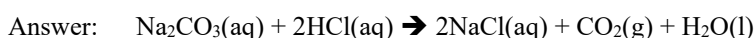
Mole of alkali : Mole of acid = n : 1

$$0.004 / 0.002 = n : 1$$

$$n = 2 \quad (\text{Acid X is a dibasic acid})$$

Example 5 (Calculate the water of crystallization from molar mass)

An experiment was performed to determine the number of water of crystallization, n, in a sample of hydrated sodium carbonate (Na₂CO₃ • nH₂O). 2.86 g of the sample was dissolved completely in about 50 cm³ of deionized water in a conical flask. The solution obtained was alkaline and was immediately titrated with 0.58 M HCl(aq). It required 34.50 cm³ of the acid to reach the end point. Calculate the number of water of crystallization, n. (Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)



$$^n\text{HCl} = (34.5/1000) \times 0.58 = 0.02$$

$$^n\text{Na}_2\text{CO}_3 \text{ in the solution} = 0.02 \times 1/2 = 0.01$$



$$^n\text{Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O} \text{ in 2.86 g sample} = ^n\text{Na}_2\text{CO}_3 = 0.01$$

$$\text{Molar mass of Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O} = \text{Mass} / \text{Mole} = 2.86/0.01 = 286$$

$$286 = 23 \times 2 + 12 + 16 \times 3 + n(1 \times 2 + 16)$$

$$n = 10$$

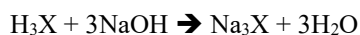
Intensive notes (Topic 4: Acids and bases)

Example 6 (Calculate the percentage by mass of a sample)

Citric acid is a tribasic acid found in lemon. A solid sample contained citric acid and other soluble inert substances. 1.65 g of the sample was dissolved in deionized water and diluted to 250.0 cm³. After that, 25.00 cm³ of the diluted solution was withdrawn and titrated with 0.123 M NaOH(aq). 18.45 cm³ of the NaOH(aq) was required to reach the end point. Calculate the percentage by mass of citric acid in the solid sample. (You may let H₃X be the formula of citric acid)

(Molar mass of citric acid = 192.0 g)

Answer:



$$n_{\text{NaOH}} = (18.45/1000) \times 0.123 = 0.00227$$

$$n_{\text{H}_3\text{X}} \text{ in 1.65 g sample} = 0.00227 \times 1/3 \times 250/25 = 7.56 \times 10^{-3}$$

$$\text{Percentage by mass} = (7.56 \times 10^{-3})(192) / 1.65 = 88.0\%$$

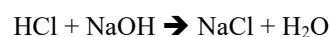
Example 7 (Back titration)

The percentage by mass of calcium carbonate in a sample of eggshell was determined. 0.204 g of the eggshell sample was put into a conical flask. After that, 25.00 cm³ of 0.200 M HCl(aq) was added. The mixture was heated for 15 minutes. After cooling down, the mixture was titrated with 0.102 M NaOH(aq). 16.85 cm³ of NaOH(aq) was needed to reach the end point of titration. Calculate the percentage by mass of calcium carbonate in the sample.

(Molar mass of calcium carbonate = 100.1 g)

Answer:

$$\text{Total } n_{\text{HCl}} \text{ used} = (25/1000) \times 0.2 = 5 \times 10^{-3}$$



$$n_{\text{NaOH}} = (16.85/1000) \times 0.102 = 1.72 \times 10^{-3} = \text{Excess } n_{\text{HCl}}$$



$$n_{\text{HCl}} \text{ reacted with CaCO}_3 = 5 \times 10^{-3} - 1.72 \times 10^{-3} = 3.28 \times 10^{-3}$$

$$n_{\text{CaCO}_3} \text{ in the sample} = 3.28 \times 10^{-3} \times 1/2 = 1.64 \times 10^{-3}$$

$$\text{Percentage by mass} = (1.64 \times 10^{-3} \times 100.1) / 0.204 = 80.5\%$$

- For example 7, we assume only CaCO₃ reacted with HCl / The reaction between CaCO₃ and HCl was complete

Example 8 (Calculate the molar mass)

To identify an unknown water-soluble metal carbonate with the chemical formula M₂CO₃, a student prepared a 100.0 cm³ aqueous solution of the carbonate using 1.14 g of the sample. 10.0 cm³ portions of the solution was titrated with 0.085 M HCl(aq). The mean titre was 25.30 cm³. Based on the experimental results, deduce what M is.

(Relative atomic masses: C = 12.0, O = 16.0)

Answer:



$$n_{\text{HCl}} = (25.3/1000) \times 0.085 = 0.00215$$

$$n_{\text{M}_2\text{CO}_3} \text{ in 1.14 g sample} = 0.00215 \times 1/2 \times 100/10 = 0.01075$$

$$\text{Molar mass of M}_2\text{CO}_3 = \text{Mass} / \text{Mole} = 1.14/0.01075 = 106 \text{ g}$$

$$106 = 2M + 12 + 16 \times 3$$

$$M = 23 \quad (\text{M should be sodium})$$

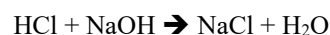
Example 9 (Back titration)

To identify an unknown metal carbonate with the chemical formula MCO₃, 25.0 cm³ of 1.50 M HCl(aq) is added to 1.29 g of MCO₃(s). The excess HCl(aq) is then back titrated with 0.300 M NaOH(aq). The mean titre was 23.00 cm³. Based on the experimental results, deduce what M is.

(Relative atomic masses: C = 12.0, O = 16.0)

Answer:

$$\text{Total } n_{\text{HCl}} \text{ used} = (25/1000) \times 1.5 = 0.0375$$



$$n_{\text{NaOH}} = (23.00/1000) \times 0.300 = 0.0069 = \text{Excess } n_{\text{HCl}}$$



$$n_{\text{HCl}} \text{ reacted with MCO}_3 = 0.0375 - 0.0069 = 0.0306$$

$$n_{\text{MCO}_3} \text{ in 1.29 g sample} = 0.0306 \times 1/2 = 0.0153$$

$$\text{Molar mass of MCO}_3 = \text{Mass} / \text{Mole} = 1.29/0.0153 = 84.3 \text{ g}$$

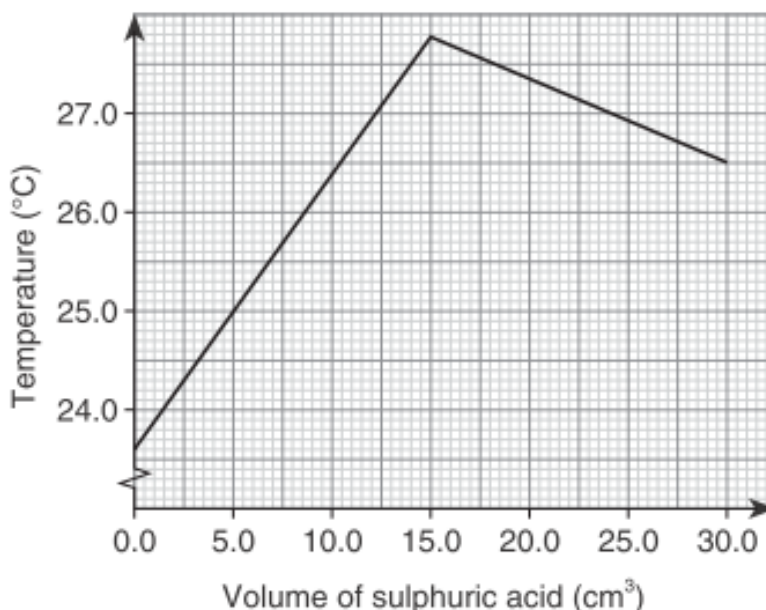
$$84.3 = M + 12 + 16 \times 3$$

$$M = 24.1 \quad (\text{M should be magnesium})$$

- For example 7 and 9, direct titration is NOT used in the experiment as CaCO₃ / MCO₃ are insoluble in water.

Example 10 (Temperature change in a titration curve)

In an experiment, 10.0 cm^3 of NaOH(aq) is placed in an expanded polystyrene cup, and the initial temperature of the alkali is measured using a thermometer. After that, 0.10 M of $\text{H}_2\text{SO}_4\text{(aq)}$ is added to the alkali, 1.00 cm^3 at a time. The temperature of the reaction mixture is recorded immediately after each addition of the acid. The following graph is obtained.



- (a) Explain why NaOH(aq) should be placed in an expanded polystyrene cup.

Answer: To reduce the heat loss to the surrounding

- (b) Explain the temperature change in the above titration curve.

Answer: The temperature of the mixture rises as the reaction between NaOH(aq) and $\text{H}_2\text{SO}_4\text{(aq)}$ is exothermic.

The temperature of the mixture reaches maximum at equivalence point (i.e. 15.0 cm^3).

The temperature of the mixture then falls due to the addition of excess H_2SO_4 .

- (c) Calculate the molarity of NaOH(aq) used.

Answer: $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$

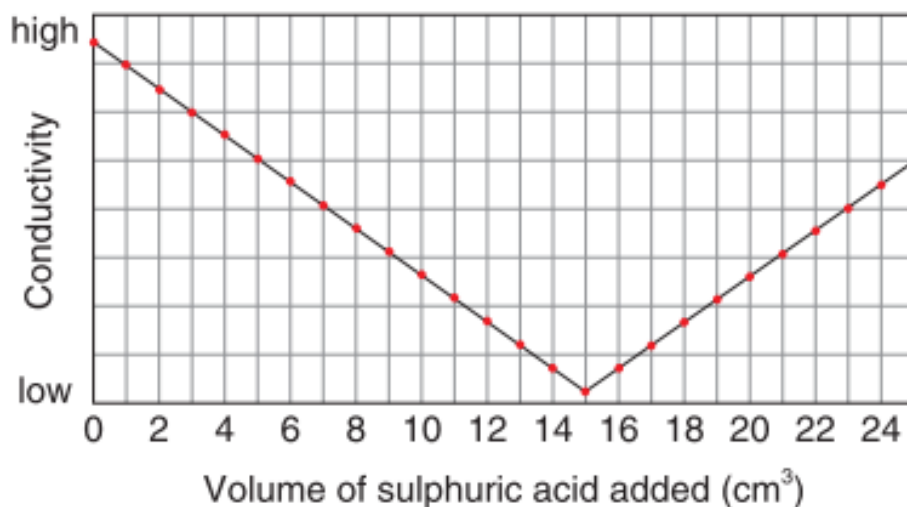
Mole of $\text{H}_2\text{SO}_4 = (15/1000) \times 0.1 = 0.0015$

Mole of $\text{NaOH} = 0.0015 \times 2 = 0.003$

Concentration of $\text{NaOH} = 0.003 / (10/1000) = 0.3 \text{ M}$

Example 11 (Conductivity change in a titration curve)

In an experiment, 25.0 cm^3 of $\text{Ba}(\text{OH})_2(\text{aq})$ is placed in an expanded polystyrene cup, and the initial electrical conductivity of the alkali is measured. After that, 0.10 M of $\text{H}_2\text{SO}_4(\text{aq})$ is added to the alkali, 1.00 cm^3 at a time. The conductivity of the reaction mixture is recorded immediately after each addition of the acid. The following graph is obtained.



- (a) Explain the electrical conductivity change in the above titration curve.

Answer: The conductivity of the mixture decreases. It is because:

- $\text{Ba}^{2+}(\text{aq})$ undergoes precipitation with $\text{SO}_4^{2-}(\text{aq})$ to give $\text{BaSO}_4(\text{s})$
- $\text{OH}^-(\text{aq})$ undergoes neutralization with $\text{H}^+(\text{aq})$ to give $\text{H}_2\text{O}(\text{l})$

Concentration of mobile ions in $\text{Ba}(\text{OH})_2(\text{aq})$ decreases.

The conductivity of the mixture reaches minimum at equivalence point (i.e. 15.0 cm^3).

The conductivity of the mixture then increases due to the addition of excess $\text{H}_2\text{SO}_4(\text{aq})$ which contains $\text{H}^+(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$ for conducting electricity

- (b) Calculate the molarity of $\text{NaOH}(\text{aq})$ used.

Answer: $\text{Ba}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{H}_2\text{O}$

Mole of $\text{H}_2\text{SO}_4 = (15/1000) \times 0.1 = 0.0015$

Mole of $\text{Ba}(\text{OH})_2 = 0.0015$

Concentration of $\text{Ba}(\text{OH})_2 = 0.0015 / (25/1000) = 0.06 \text{ M}$